



Security Operations, Log Management, SIEM, Tools & Incident Response

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DATE : 01-April-2026

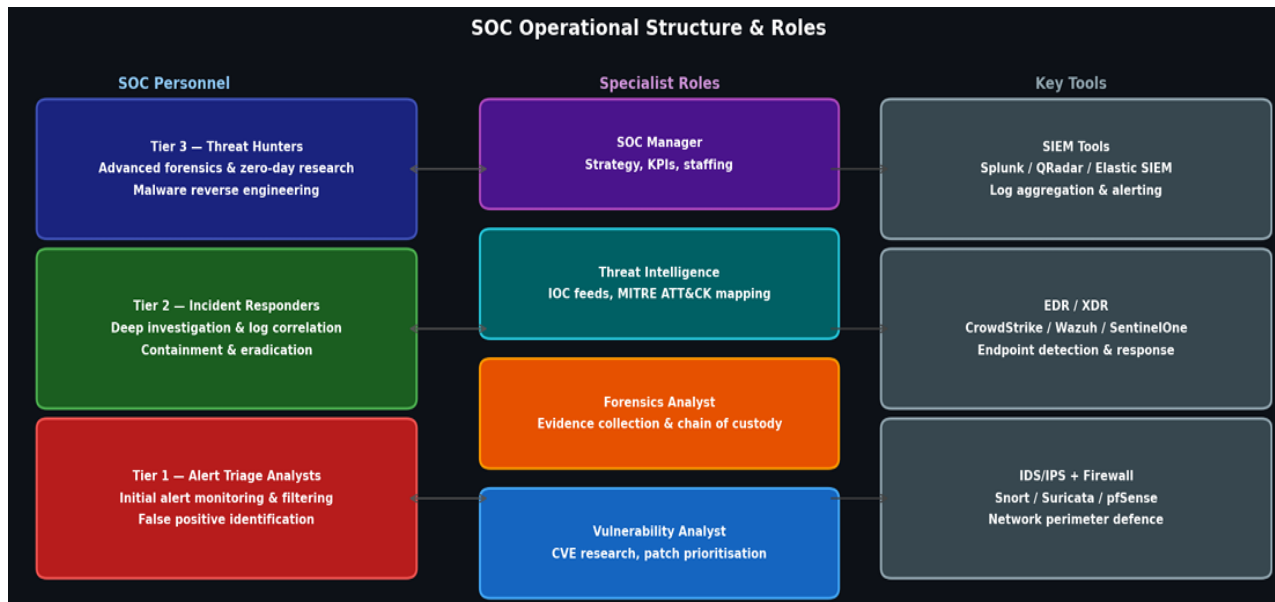
1. SOC Fundamentals & Operations

A Security Operations Centre (SOC) is a centralised team responsible for continuously monitoring, detecting, analysing, and responding to cybersecurity incidents. The three core functions are:

- **Proactive Threat Detection:** Monitor systems 24/7 for anomalies, indicators of compromise (IOCs), and policy violations before they escalate.
- **Incident Response:** When a threat is confirmed, contain it, eradicate the root cause, recover affected systems, and document lessons learned.
- **Continuous Monitoring:** Aggregate logs from all sources (endpoints, network, cloud), correlate them in a SIEM, and generate alerts using rule-based and ML-based detection.

SOC Tier Structure & Roles

Role	Responsibilities	Key Focus
Tier 1 Analyst (L1)	Monitor alerts, initial triage, escalate issues	First line of defense
Tier 2 Analyst (L2)	Investigate alerts, incident response, root cause analysis	Deep investigation
Tier 3 Analyst (L3)	Handle advanced threats, create detection rules	Expert-level analysis
SOC Manager	Manage team, define policies, ensure SLA compliance	Leadership & strategy
Threat Hunter	Proactively search for hidden threats	Attack detection before alerts



2. Security Monitoring Basics

Objectives:

- 1. Detect anomalies:** Unusual traffic volumes, out-of-hours logins, processes running from temp directories, abnormal outbound connections.
- 2. Unauthorized access:** Failed login spikes (Event ID 4625), privilege escalation (Event ID 4672), new admin account creation (4720).
- 3. Policy violations:** USB device insertion, cloud storage uploads, software installation outside change windows.

SIEM — Elastic Stack Setup (Ubuntu)

Step 1: Import the Elasticsearch PGP key

Command:

- `wget -qO - https://artifacts.elastic.co/GPG-KEY-elasticsearch | sudo gpg --dearmor -o /usr/share/keyrings/elasticsearch-keyring.gpg`



```
root@bhanu-VirtualBox:~$ sudo su
[sudo] password for bhanu:
root@bhanu-VirtualBox:/home/bhanu# wget -q0 - https://artifacts.elastic.co/GPG-KEY-elasticsearch
root@bhanu-VirtualBox:/home/bhanu# sudo apt-get install apt-transport-https
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
apt-transport-https is already the newest version (2.8.3).
The following package was automatically installed and is no longer required:
  libllvm19
Use 'sudo apt autoremove' to remove it.
0 upgraded, 0 newly installed, 0 to remove and 0 not upgraded.
```

Step 2: Install Elasticsearch

Install from the APT repository

Command:

- `sudo apt-get install apt-transport-https`
- `echo "deb [signed-by=/usr/share/keyrings/elasticsearch-keyring.gpg] https://artifacts.elastic.co/packages/9.x/apt stable main" | sudo tee /etc/apt/sources.list.d/elastic-9.x.list`

```
root@bhanu-VirtualBox:/home/bhanu# echo "deb [signed-by=/usr/share/keyrings/elasticsearch-keyring.gpg] https://artifacts.elastic.co/packages/9.x/apt stable main" | sudo tee /etc/apt/sources.list.d/elastic-9.x.list
deb [signed-by=/usr/share/keyrings/elasticsearch-keyring.gpg] https://artifacts.elastic.co/packages/9.x/apt stable main
root@bhanu-VirtualBox:/home/bhanu# sudo apt-get update && sudo apt-get install elasticsearch
Hit:1 http://in.archive.ubuntu.com/ubuntu noble InRelease
Get:2 http://security.ubuntu.com/ubuntu noble-security InRelease [126 kB]
Get:3 http://in.archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]
Get:4 http://in.archive.ubuntu.com/ubuntu noble-backports InRelease [126 kB]
Get:5 https://artifacts.elastic.co/packages/9.x/apt stable InRelease [3,249 B]
Get:6 http://in.archive.ubuntu.com/ubuntu noble-updates/main amd64 Packages [1,870 kB]
Get:7 http://security.ubuntu.com/ubuntu noble-security/main amd64 Packages [1,551 kB]
Get:8 http://in.archive.ubuntu.com/ubuntu noble-updates/main Translation-en [342 kB]
Get:9 http://in.archive.ubuntu.com/ubuntu noble-updates/main amd64 Components [177 kB]
Get:10 http://in.archive.ubuntu.com/ubuntu noble-updates/main amd64 c-n-f Metadata [16.9 kB]
Get:11 http://in.archive.ubuntu.com/ubuntu noble-updates/restricted amd64 Packages [2,891 kB]
Get:12 http://in.archive.ubuntu.com/ubuntu noble-updates/restricted Translation-en [672 kB]
Get:13 http://security.ubuntu.com/ubuntu noble-security/main Translation-en [250 kB]
Get:14 http://security.ubuntu.com/ubuntu noble-security/main amd64 Components [21.5 kB]
Get:15 http://in.archive.ubuntu.com/ubuntu noble-updates/restricted amd64 Components [212 B]
Get:16 http://security.ubuntu.com/ubuntu noble-security/main amd64 c-n-f Metadata [10.7 kB]
```



- To install the latest version of Elasticsearch 9.3.2 package, enter:

Command:

- `sudo apt-get update && sudo apt-get install elasticsearch`

```
root@bhanu-VirtualBox:/home/bhanu# sudo apt-get update && sudo apt-get install elasticsearch
Hit:1 http://in.archive.ubuntu.com/ubuntu noble InRelease
Get:2 http://security.ubuntu.com/ubuntu noble-security InRelease [126 kB]
Get:3 http://in.archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]
Get:4 http://in.archive.ubuntu.com/ubuntu noble-backports InRelease [126 kB]
Get:5 https://artifacts.elastic.co/packages/9.x/apt stable InRelease [3,249 B]
Get:6 http://in.archive.ubuntu.com/ubuntu noble-updates/main amd64 Packages [1,870 kB]
Get:7 http://security.ubuntu.com/ubuntu noble-security/main amd64 Packages [1,551 kB]
Get:8 http://in.archive.ubuntu.com/ubuntu noble-updates/main Translation-en [342 kB]
Get:9 http://in.archive.ubuntu.com/ubuntu noble-updates/main amd64 Components [177 kB]
Get:10 http://in.archive.ubuntu.com/ubuntu noble-updates/main amd64 c-n-f Metadata [16.9 kB]
Get:11 http://in.archive.ubuntu.com/ubuntu noble-updates/restricted amd64 Packages [2,891 kB]
]
Get:12 http://in.archive.ubuntu.com/ubuntu noble-updates/restricted Translation-en [672 kB]
Get:13 http://security.ubuntu.com/ubuntu noble-security/main Translation-en [250 kB]
Get:14 http://security.ubuntu.com/ubuntu noble-security/main amd64 Components [21.5 kB]
Get:15 http://in.archive.ubuntu.com/ubuntu noble-updates/restricted amd64 Components [212 B]
Get:16 http://security.ubuntu.com/ubuntu noble-security/main amd64 c-n-f Metadata [10.7 kB]
Get:17 http://security.ubuntu.com/ubuntu noble-security/restricted amd64 Components [212 B]
Get:18 http://security.ubuntu.com/ubuntu noble-security/universe amd64 Packages [1,166 kB]
Get:19 http://in.archive.ubuntu.com/ubuntu noble-updates/universe amd64 Packages [1,665 kB]
Get:20 http://security.ubuntu.com/ubuntu noble-security/universe amd64 Components [74.2 kB]
Get:21 http://security.ubuntu.com/ubuntu noble-security/universe amd64 c-n-f Metadata [22.8
kB]
```

Run Elasticsearch with systemd

- `sudo /bin/systemctl daemon-reload`
- `sudo /bin/systemctl enable elasticsearch.service`

```
root@bhanu-VirtualBox:/home/bhanu
bhanu@bhanu-VirtualBox:~$ sudo su
[sudo] password for bhanu:
root@bhanu-VirtualBox:/home/bhanu# sudo /bin/systemctl daemon-reload
root@bhanu-VirtualBox:/home/bhanu# sudo /bin/systemctl enable elasticsearch.service
Created symlink /etc/systemd/system/multi-user.target.wants/elasticsearch.service → /usr/lib/
systemd/system/elasticsearch.service.
root@bhanu-VirtualBox:/home/bhanu# sudo systemctl start elasticsearch.service
```



Elasticsearch can be started and stopped as follows:

- `sudo systemctl start elasticsearch.service`
- `sudo systemctl stop elasticsearch.service`

```
root@bhanu-VirtualBox:/home/bhanu# sudo systemctl start elasticsearch.service
```

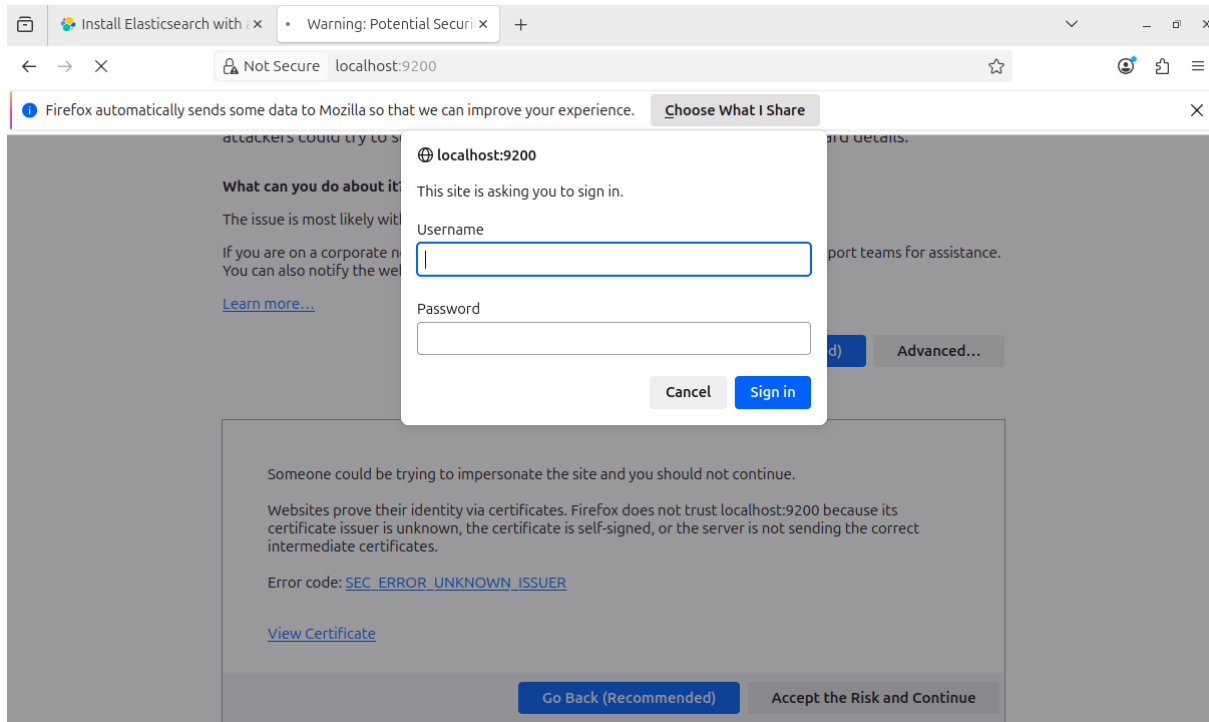
To check the status :

- `sudo systemctl status elasticsearch.service`

```
root@bhanu-VirtualBox:/home/bhanu# sudo systemctl enable elasticsearch.service
Created symlink /etc/systemd/system/multi-user.target.wants/elasticsearch.service → /usr/lib/systemd/system/elasticsearch.service.
root@bhanu-VirtualBox:/home/bhanu# sudo systemctl start elasticsearch.service
root@bhanu-VirtualBox:/home/bhanu# sudo systemctl status elasticsearch.service
● elasticsearch.service - Elasticsearch
   Loaded: loaded (/usr/lib/systemd/system/elasticsearch.service; enabled; preset: enabled)
   Active: active (running) since Wed 2026-04-01 06:50:49 MDT; 1min 37s ago
     Docs: https://www.elastic.co
  Main PID: 4211 (java)
    Tasks: 100 (limit: 4597)
   Memory: 2.3G (peak: 2.3G)
      CPU: 4min 56.404s
   CGroup: /system.slice/elasticsearch.service
           └─4211 /usr/share/elasticsearch/jdk/bin/java -Xms4m -Xmx64m -XX:+UseSerialGC -D>
           └─4274 /usr/share/elasticsearch/jdk/bin/java -Des.networkaddress.cache.ttl=60 ->
ce,safepoint:file=gc.log:utctime,level,pid,tags:filecount=32,filesize=64m" -Xms1907m -Xmx190>
           └─4375 /usr/share/elasticsearch/modules/x-pack-ml/platform/linux-x86_64/bin/con>

Apr 01 06:46:53 bhanu-VirtualBox systemd[1]: Starting elasticsearch.service - Elasticsearch.>
Apr 01 06:47:37 bhanu-VirtualBox systemd-entrypoint[4211]: WARNING: Unknown module: jdk.inte>
Apr 01 06:47:37 bhanu-VirtualBox systemd-entrypoint[4211]: WARNING: Unknown module: org.elas>
Apr 01 06:50:49 bhanu-VirtualBox systemd[1]: Started elasticsearch.service - Elasticsearch.
lines 1-18/18 (END)
```

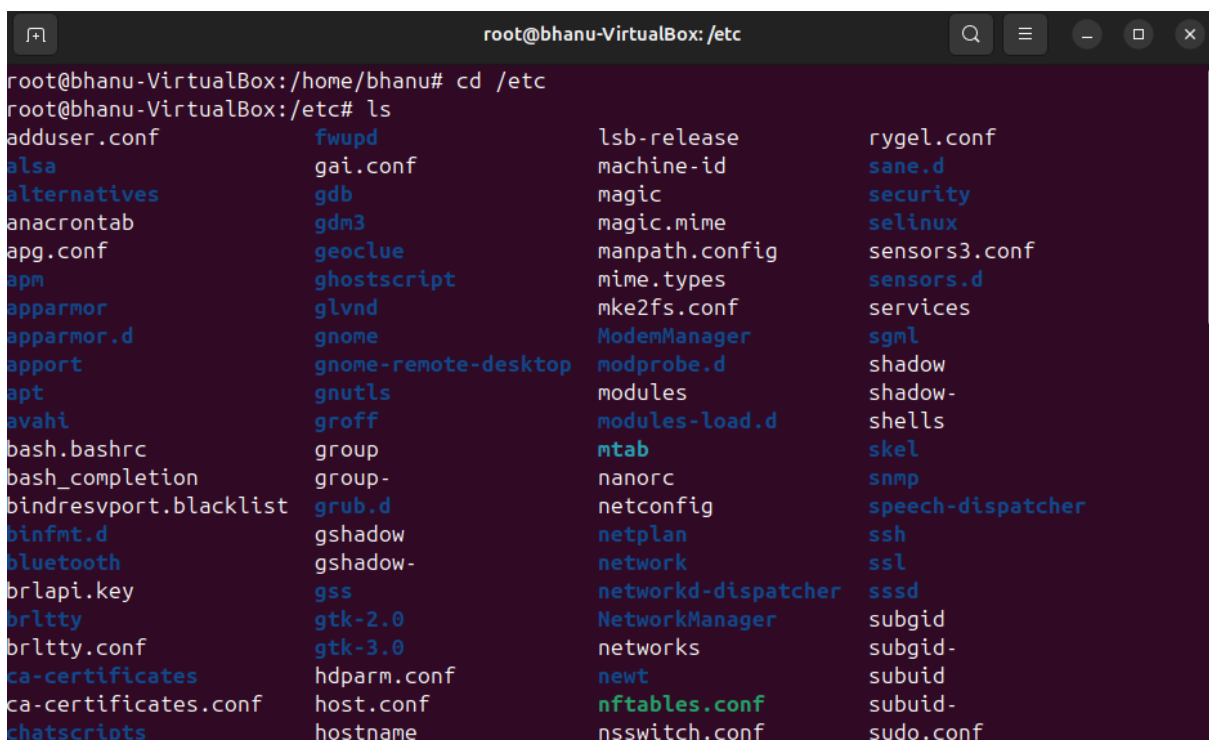
- Here we can see the status is active
- Default port is 9200



- Here the site is asking username and password
- We can also disable the username and password

To disable

- cd /etc
- ls





```
root@bhanu-VirtualBox:/etc# sudo chmod 777 elasticsearch/
root@bhanu-VirtualBox:/etc# cd elasticsearch/
root@bhanu-VirtualBox:/etc/elasticsearch# ls
certs                    elasticsearch.yml        log4j2.properties      users
elasticsearch.keystore   jvm.options             role_mapping.yml        users_roles
elasticsearch-plugins.example.yml  jvm.options.d           roles.yml
root@bhanu-VirtualBox:/etc/elasticsearch# sudo chmod 777 elasticsearch.yml
root@bhanu-VirtualBox:/etc/elasticsearch# sudo nano elasticsearch.yml
```

Change this to true to false

- CIt +x
- Y = save
- Enter

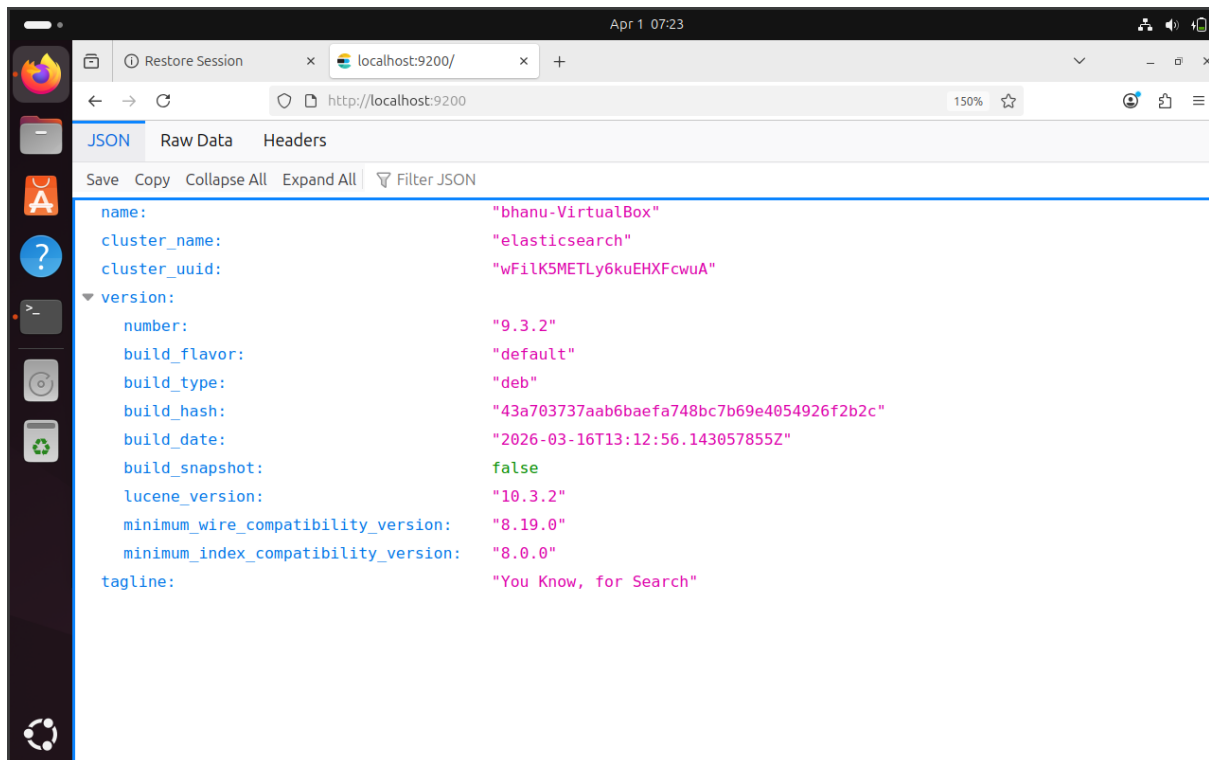
After restart the service to start the elastic

Command :

- sudo systemctl restart elasticsearch.service

Again check the status of the service with :

- sudo systemctl status elasticsearch.service

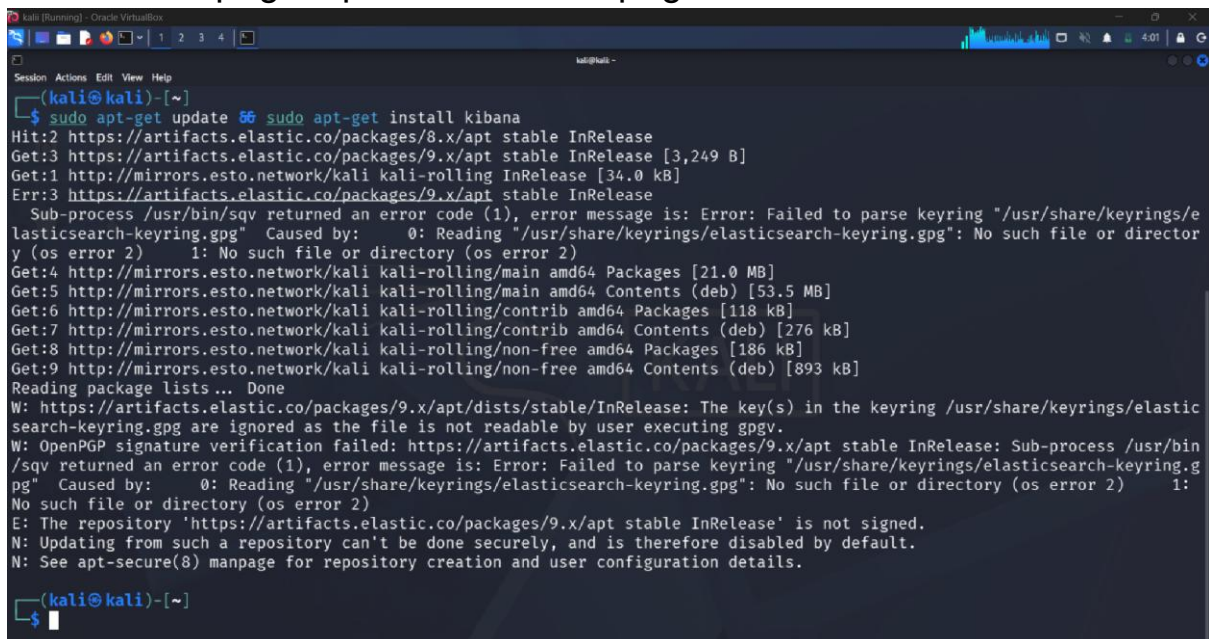


Install Kibana (Kali)

Install the Kibana Debian package

Command:

- `sudo apt-get update && sudo apt-get install kibana`



To configure Kibana to start automatically when the system starts, run the following commands:



- sudo /bin/systemctl daemon-reload
- sudo /bin/systemctl enable kibana.service

```
(kali@kali)-[~]
$ sudo systemctl start kibana.service

(kali@kali)-[~]
$ sudo systemctl status kibana.service
● kibana.service - Kibana
   Loaded: loaded (/usr/lib/systemd/system/kibana.service; enabled; preset: disabled)
   Active: active (running) since Thu 2026-04-02 03:57:14 EDT, 5min ago
   Invocation: e70d2adf57ed41cd5e91c2fa4679b60
     Docs: https://www.elastic.co
   Main PID: 693 (node)
     Tasks: 11 (limit: 4456)
  Memory: 325.1M (peak: 582.5M, swap: 77.7M, swap peak: 77.7M)
     CPU: 36.817s
   CGroup: /system.slice/kibana.service
           └─693 /usr/share/kibana/bin/../../node/glibc-217/bin/node /usr/share/kibana/bin/../../src/cli/kibana/dist

Apr 02 03:57:56 kali kibana[693]: Native global console methods have been overridden in production environment.
Apr 02 03:58:28 kali kibana[693]: [2026-04-02T03:58:27.798-04:00][INFO ][root] Kibana is starting
Apr 02 03:58:28 kali kibana[693]: [2026-04-02T03:58:28.765-04:00][INFO ][node] Kibana process configured with roles: [backgro
Apr 02 03:59:32 kali kibana[693]: [2026-04-02T03:59:32.240-04:00][INFO ][plugins-service] The following plugins are disabled:
Apr 02 03:59:32 kali kibana[693]: [2026-04-02T03:59:32.739-04:00][INFO ][http.server.Preboot] http server running at http://l
Apr 02 03:59:33 kali kibana[693]: [2026-04-02T03:59:33.107-04:00][INFO ][plugins-system.preboot] Setting up [1] plugins: [int
Apr 02 03:59:33 kali kibana[693]: [2026-04-02T03:59:33.218-04:00][INFO ][preboot] "interactiveSetup" plugin is holding setup:
Apr 02 03:59:33 kali kibana[693]: [2026-04-02T03:59:33.269-04:00][INFO ][root] Holding setup until preboot stage is completed.
Apr 02 03:59:42 kali kibana[693]: i Kibana has not been configured.
Apr 02 03:59:42 kali kibana[693]: Go to http://localhost:5601/?code=580464 to get started.
lines 1-22/22 (END)
```

Kibana can be started and stopped :

- sudo systemctl start kibana.service
- sudo systemctl stop kibana.service

Kibana status check :

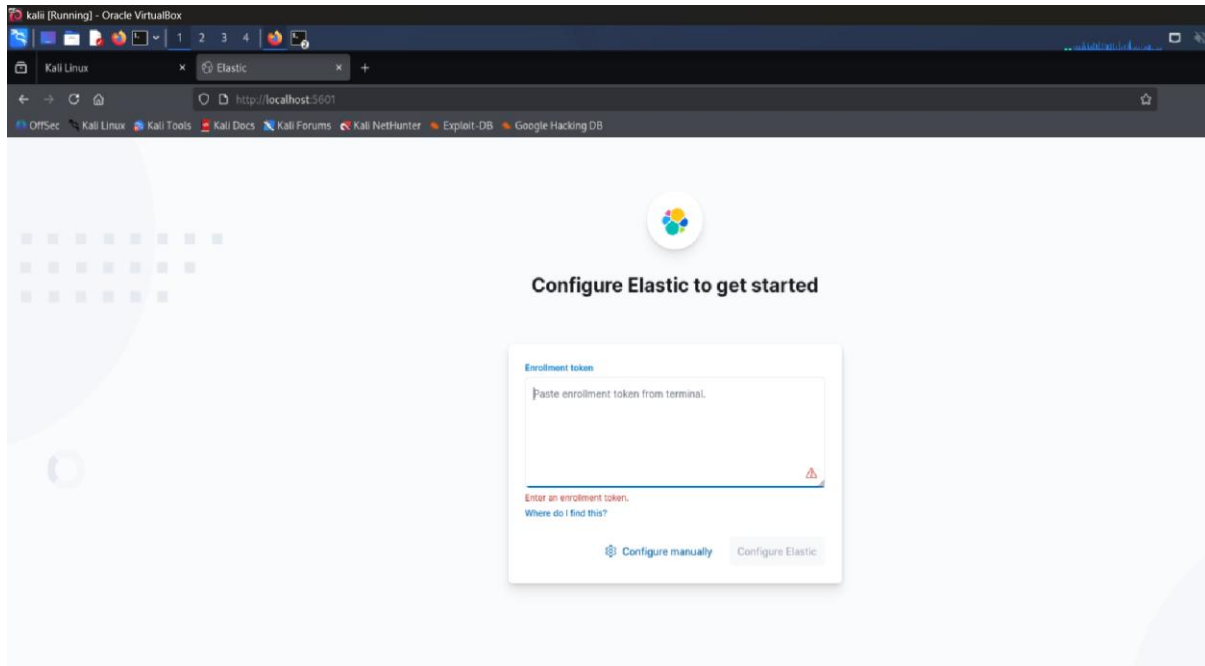
- sudo systemctl status kibana.service

```
(kali@kali)-[~]
$ sudo systemctl status kibana.service
● kibana.service - Kibana
   Loaded: loaded (/usr/lib/systemd/system/kibana.service; enabled; preset: disabled)
   Active: active (running) since Thu 2026-04-02 03:57:14 EDT, 5min ago
   Invocation: e70d2adf57ed41cd5e91c2fa4679b60
     Docs: https://www.elastic.co
   Main PID: 693 (node)
     Tasks: 11 (limit: 4456)
  Memory: 325.1M (peak: 582.5M, swap: 77.7M, swap peak: 77.7M)
     CPU: 36.817s
   CGroup: /system.slice/kibana.service
           └─693 /usr/share/kibana/bin/../../node/glibc-217/bin/node /usr/share/kibana/bin/../../src/cli/kibana/dist

Apr 02 03:57:56 kali kibana[693]: Native global console methods have been overridden in production environment.
Apr 02 03:58:28 kali kibana[693]: [2026-04-02T03:58:27.798-04:00][INFO ][root] Kibana is starting
Apr 02 03:58:28 kali kibana[693]: [2026-04-02T03:58:28.765-04:00][INFO ][node] Kibana process configured with roles: [backgro
Apr 02 03:59:32 kali kibana[693]: [2026-04-02T03:59:32.240-04:00][INFO ][plugins-service] The following plugins are disabled:
Apr 02 03:59:32 kali kibana[693]: [2026-04-02T03:59:32.739-04:00][INFO ][http.server.Preboot] http server running at http://l
Apr 02 03:59:33 kali kibana[693]: [2026-04-02T03:59:33.107-04:00][INFO ][plugins-system.preboot] Setting up [1] plugins: [int
Apr 02 03:59:33 kali kibana[693]: [2026-04-02T03:59:33.218-04:00][INFO ][preboot] "interactiveSetup" plugin is holding setup:
Apr 02 03:59:33 kali kibana[693]: [2026-04-02T03:59:33.269-04:00][INFO ][root] Holding setup until preboot stage is completed.
Apr 02 03:59:42 kali kibana[693]: i Kibana has not been configured.
Apr 02 03:59:42 kali kibana[693]: Go to http://localhost:5601/?code=580464 to get started.
lines 1-22/22 (END)
```

Run the kibana in the browser of kali

- <http://localhost:5601>



3. Log Management Fundamentals

1. Log Lifecycle

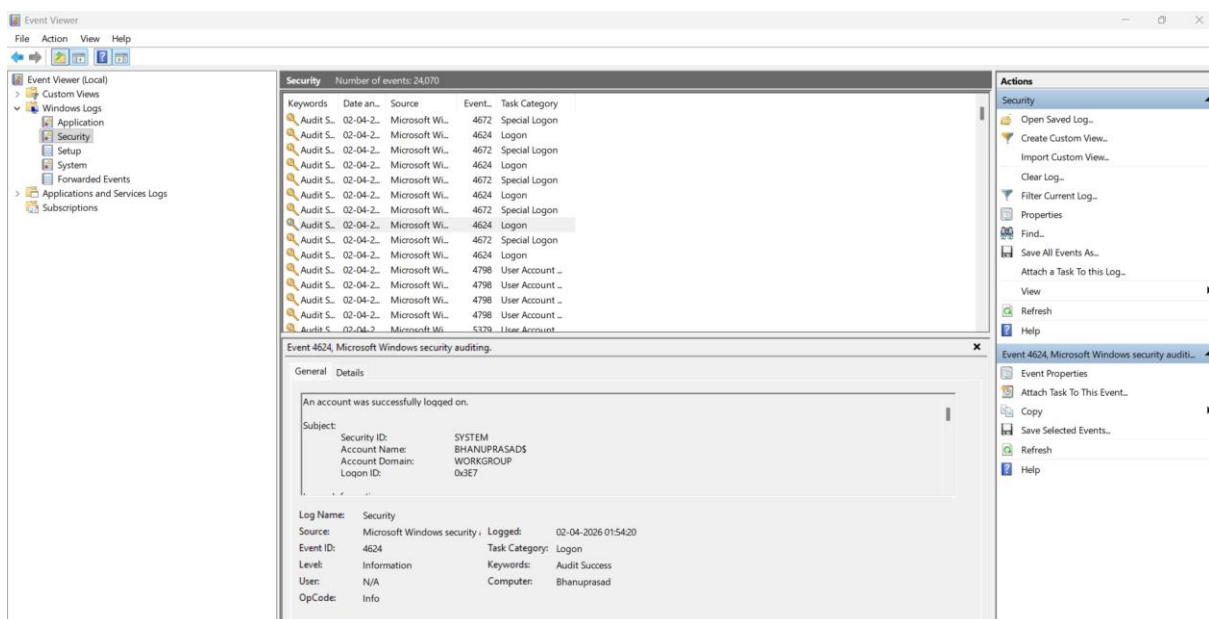
Stage	Description	Example
Collection	Gathering logs from systems/devices	Syslog from Linux, Event Logs from Windows
Normalization	Converting logs into standard format	JSON, CEF
Storage	Saving logs in centralized system	Elasticsearch
Retention	Keeping logs for a defined period	30 days / 90 days
Analysis	Searching & detecting threats	Queries in Kibana



Common Log Types :

Log Type	Source	Use Case
Windows Event Logs	Windows OS	Detect failed logins (Event ID 4625)
Syslog	Linux systems	Monitor system activity
HTTP Server Logs	Apache/Nginx	Track web traffic, attacks
Firewall Logs	Network devices	Detect suspicious connections

- View system logs of our main window in the event viewer



Where to See All Log Types in Your System (Ubuntu)

- All system logs are stored in:



- ls /var/log

```
bhanu@bhanu-VirtualBox: /var/log
bhanu@bhanu-VirtualBox:~$ cd /var/log/
bhanu@bhanu-VirtualBox:/var/log$ ls
alternatives.log      cups                  gpu-manager.log      speech-dispatcher
appport.log           cups-browsed         hp                   sssd
apt                   dist-upgrade         installer            syslog
auth.log              dmesg                journal             sysstat
boot.log              dpkg.log             kern.log             ubuntu-advantage-apt-hook.log
bootstrap.log         elasticsearch         lastlog              unattended-upgrades
btmtp                 faillog              openvpn              wtmp
cloud-init.log        fontconfig.log       private
cloud-init-output.log gdm3                 README
```

Installing logstash on (Ubuntu)

Command:

- sudo apt-get update && sudo apt-get install logstash

```
~$ sudo apt-get install logstash
Reading package lists... Done
Building dependency tree
Reading state information... Done
logstash is already the newest version (1:7.10.0-1).
0 upgraded, 0 newly installed, 0 to remove and 9 not upgraded.
~$ sudo systemctl status logstash.service
logstash.service - logstash
Loaded: loaded (/etc/systemd/system/logstash.service; disabled; vendor preset: enabled)
Active: inactive (dead)

Dec 05 18:28:27 anubook logstash[69245]: [2020-12-05T18:28:27.077][INFO ][logstash.inputs.tcp ][main][1b74b7e0bf8bbdff4adef268f35e75f7cbc952da37]
Dec 05 18:28:27 anubook logstash[69245]: [2020-12-05T18:28:27.237][INFO ][logstash.agent ][main] Pipelines running {:count=>1, :running_pipelines=>1}
Dec 05 18:28:27 anubook logstash[69245]: [2020-12-05T18:28:27.926][INFO ][logstash.agent ][main] Successfully started Logstash API endpoint {:port=>5044}
Dec 05 18:50:41 anubook logstash[69245]: [2020-12-05T18:50:41.369][WARN ][logstash.codecs.line ][main][1b74b7e0bf8bbdff4adef268f35e75f7cbc952da37]
Dec 05 18:51:10 anubook systemd[1]: Stopping logstash...
Dec 05 18:51:10 anubook logstash[69245]: [2020-12-05T18:51:10.530][WARN ][logstash.runner ][main] SIGTERM received. Shutting down.
Dec 05 18:51:15 anubook logstash[69245]: [2020-12-05T18:51:15.734][INFO ][logstash.javapipeline ][main] Pipeline terminated {"pipeline.id"=>"main"}
Dec 05 18:51:16 anubook logstash[69245]: [2020-12-05T18:51:16.764][INFO ][logstash.runner ][main] Logstash shut down.
Dec 05 18:51:16 anubook systemd[1]: logstash.service: Succeeded.
Dec 05 18:51:16 anubook systemd[1]: Stopped logstash.
```

To start the logstash :

Command:

- sudo systemctl start logstash.service
- sudo systemctl status logstash.service

```
~$ sudo systemctl start logstash.service
~$ sudo systemctl status logstash.service
logstash.service - logstash
Loaded: loaded (/etc/systemd/system/logstash.service; disabled; vendor preset: enabled)
Active: active (running) since Sat 2020-12-05 19:08:40 IST; 2s ago
Main PID: 77727 (java)
Tasks: 14 (limit: 9353)
Memory: 217.2M
CGroup: /system.slice/logstash.service
└─77727 /usr/share/logstash/jdk/bin/java -Xms1g -Xmx1g -XX:+UseConcMarkSweepGC -XX:CMSInitiatingOccupancyFraction=75 -XX:+UseCMSInit
Dec 05 19:08:40 anubook systemd[1]: Started logstash.
Dec 05 19:08:40 anubook logstash[77727]: Using bundled JDK: /usr/share/logstash/jdk
Dec 05 19:08:40 anubook logstash[77727]: OpenJDK 64-Bit Server VM warning: Option UseConcMarkSweepGC was deprecated in version 9.0 and will likely be removed in a future release.
```

- Configure logstash



```
1 input {
2   tcp {
3     port => 5000
4   }
5 }
6
7 output {
8   elasticsearch {
9     hosts => ["elasticsearch:9200"]
10    index => "hello-logstash"
11  }
12 }
```

```
→ ~ cd /etc/logstash/conf.d
→ conf.d ls
→ conf.d sudo vim port.conf
→ conf.d telnet localhost 5000
Trying 127.0.0.1...
Connected to localhost.
Escape character is '^]'.
hello
```

Console Search Profiler Grok Debugger Painless Lab **BETA**

History Settings Help

404 - Not Found 59 ms

```
940
941
942
943
944
945
946 GET /_cat/indices
947
948 GET hello-logstash/_search
949 {
950   "query": {
951     "match_all": {}
952   }
953 }
954
955
956
957
958
959
960
961
962
963
964
965
966
967
968
969
970
971
972
973
```

```
1- {
2-   "error": {
3-     "root_cause": [
4-       {
5-         "type": "index_not_found_exception",
6-         "reason": "no such index [hello-logstash]",
7-         "resource.type": "index_or_alias",
8-         "resource.id": "hello-logstash",
9-         "index_uuid": "_na_",
10        "index": "hello-logstash"
11      },
12      {
13        "type": "index_not_found_exception",
14        "reason": "no such index [hello-logstash]",
15        "resource.type": "index_or_alias",
16        "resource.id": "hello-logstash",
17        "index_uuid": "_na_",
18        "index": "hello-logstash"
19      }
20    ],
21    "status": 404
22  }
```

Console Search Profiler Grok Debugger Painless Lab **BETA**

History Settings Help

200 - OK 56 ms

```
940
941
942
943
944
945
946 GET /_cat/indices
947
948 GET hello-logstash/_search
949 {
950   "query": {
951     "match_all": {}
952   }
953 }
954
955
956
957
958
959
960
961
962
963
```

1	green	open	.kibana-event-log-7.9.3-000001	vpf5hqlVQx-DXexTHmu6Q	1	0	6	0	32.3kb	32.3kb
2	green	open	.kibana_task_manager_2	q2b99pt8TuCqzvs1uj9-kA	1	0	6	216	53.1kb	53.1kb
3	yellow	open	hello-logstash	TT8zn7NSQ56ksM5MtIohgQ	1	1	1	0	5.5kb	5.5kb
4	green	open	.apm-custom-link	TP0re3jk50-LX3q5AUhXSA	1	0	0	208b	208b	208b
5	green	open	.kibana_task_manager_1	t8s24hduRHRtC2FLXxH8Q0	1	0	6	30	46.1kb	46.1kb
6	green	open	.apm-agent-configuration	mn0eFbaj5H-jh5MinzenSYA	1	0	0	208b	208b	208b
7	green	open	.kibana-event-log-7.10.0-000001	l1ou455SY2UdsAc7Jh46g	1	0	1	0	5.6kb	5.6kb
8	green	open	.kibana_2	2Rak0MB4T8GvE6n4DyQk2A	1	0	18	3	10.4nb	10.4nb
9	green	open	.kibana_1	0HoeVw1VRKqCJVVG2AIaog	1	0	13	2	10.4nb	10.4nb
10										



```
History Settings Help 200 - OK
940
941
942
943
944
945 GET /_cat/indices
946
947
948 GET hello-logstash/_search
949 {
950   "query": {
951     "match_all": {}
952   }
953 }
954
955
956
957
958
959
960
961
962
963
964
965
966
967
968
969
1
{
  "took": 1,
  "timed_out": false,
  "_shards": {
    "total": 1,
    "successful": 1,
    "skipped": 0,
    "failed": 0
  },
  "hits": {
    "total": {
      "value": 1,
      "relation": "eq"
    },
    "max_score": 1.0,
    "hits": [
      {
        "_index": "hello-logstash",
        "_type": "_doc",
        "_id": "t3UqM3Y8GB1NX5wGtonf",
        "_score": 1.0,
        "_source": {
          "port": 39478,
          "host": "localhost",
          "@timestamp": "2020-12-05T13:50:04.890Z",
          "message": "hello"
        },
        "@version": "1"
      }
    ]
  }
}
```

4. Security Tools — Snort, Wazuh, Osquery & Nessus

Security Tools Overview

Category	Tool	Purpose
SIEM	Splunk, IBM QRadar	Log analysis, alerting
EDR	CrowdStrike Falcon	Endpoint threat detection
IDS/IPS	Snort	Detect/block network attacks
Vulnerability Scanner	Nessus	Find system weaknesses
Open-source SIEM	Wazuh	Monitoring + detection

Install Snort on Kali

Command:

- sudo apt update
- sudo apt install snort -y



```
Upgrading:
  snort snort-common snort-common-libraries snort-rules-default

Summary:
  Upgrading: 4, Installing: 0, Removing: 0, Not Upgrading: 278
  Download size: 3,110 kB
  Space needed: 2,048 B / 44.9 GB available

Get:1 http://kali.download/kali kali-rolling/main amd64 snort-common-libraries amd64 3.12.1.0-0kali1 [290 kB]
Get:2 http://kali.download/kali kali-rolling/main amd64 snort-rules-default all 3.12.1.0-0kali1 [233 kB]
Get:3 http://kali.download/kali kali-rolling/main amd64 snort amd64 3.12.1.0-0kali1 [2,457 kB]
Get:4 http://mirrors.esto.network/kali kali-rolling/main amd64 snort-common all 3.12.1.0-0kali1 [129 kB]
Fetched 3,110 kB in 3s (1,102 kB/s)
(Reading database... 582742 files and directories currently installed.)
Preparing to unpack .../snort-common-libraries_3.12.1.0-0kali1_amd64.deb...
Unpacking snort-common-libraries (3.12.1.0-0kali1) over (3.11.1.0-0kali1)...
Preparing to unpack .../snort-rules-default_3.12.1.0-0kali1_all.deb...
Unpacking snort-rules-default (3.12.1.0-0kali1) over (3.11.1.0-0kali1)...
Preparing to unpack .../snort_3.12.1.0-0kali1_amd64.deb...
Unpacking snort (3.12.1.0-0kali1) over (3.11.1.0-0kali1)...
Preparing to unpack .../snort-common_3.12.1.0-0kali1_all.deb...
Unpacking snort-common (3.12.1.0-0kali1) over (3.11.1.0-0kali1)...
Setting up snort-common (3.12.1.0-0kali1)...
Installing new version of config file /etc/snort/snort.lua...
Setting up snort-rules-default (3.12.1.0-0kali1)...
Setting up snort-common-libraries (3.12.1.0-0kali1)...
Setting up snort (3.12.1.0-0kali1)...
snort.service is a disabled or a static unit not running, not starting it.
Processing triggers for kali-menu (2026.1.5)...
Processing triggers for man-db (2.13.1-1)...
```

Add Your Rule

Command:

- `sudo nano /etc/snort/rules/local.rules`
- `alert tcp any any -> any 80 (msg:"Malicious Domain"; content:"malicious.com"; http_uri; sid:1000001;)`

```
Session Actions Edit View Help
GNU nano 8.7.1 /etc/snort/rules/local.rules
# $Id: local.rules,v 1.11 2004/07/23 20:15:44 bmc Exp $
#
# LOCAL RULES
#
# This file intentionally does not come with signatures. Put your local
# additions here.
alert tcp any any -> any any (msg:"SYN Packet Detected"; flags:S; sid:1000001;)
alert tcp any any -> any 1-1024 (msg:"Potential Port Scan"; flags:S; sid:1000002;)
alert tcp any any -> any 80 (msg:"Malicious Domain"; content:"malicious.com"; http_uri; sid:1000001;)
```

- Press CTRL + O → Enter
- Press CTRL + X

Make Sure Rule is Loaded

Open config:



- `sudo nano /etc/snort/snort.conf`

check :

- `include $RULE_PATH/local.rules`

A screenshot of the GNU nano 8.7.1 text editor. The menu bar at the top shows 'Session Actions Edit View Help'. The main text area contains the line `include $RULE_PATH/local.rules` with a cursor at the end of the line. A large, semi-transparent 'KALI' watermark is visible in the background.

Run snort:

- `sudo snort -A console -q -c /etc/snort/snort.conf -i eth0`

A screenshot of a terminal window showing the execution of the snort command. The prompt is `(kali@kali)-[~]`. The command entered is `$ sudo snort -c /etc/snort/snort.lua -i eth0`. The output shows the Snort++ version `3.12.1.0` and a list of loaded modules: `ips`, `classifications`, `references`, `binder`, `wizard`, `smtp`, `js_norm`, `file_policy`, `file_inspect`, `http2_inspect`, `ftp_server`, `ftp_data`, `active`, `alerts`, `daq`, `host_cache`, `host_tracker`, `network`, `packets`, `process`, `search_engine`, `ftp_client`, `stream`, `stream_ip`, `stream_icmp`, `stream_udp`, `stream_user`, `arp_spoof`, `back_orifice`, `dns`, `normalizer`, `rpc_decode`, and `sip`.



Test Malicious Domain

- curl <http://malicious.com>

```
Finished file loop to_client: 1

search engine (ac_bnfa)
  instances: 2
  patterns: 438
  pattern chars: 2602
  num states: 1832
  num match states: 392
  memory scale: KB
  total memory: 71.2812
  pattern memory: 19.6484
  match list memory: 28.4375
  transition memory: 22.9453
appid: MaxRss diff: 2688
appid: patterns loaded: 300

pcap DAQ configured to passive.
Commencing packet processing
Retry queue interval is: 200 ms
++ [0] wlan0
Couldn't start DAQ instance (wlan0): No such device exists (-1)
Analyzer: Failed to start DAQ instance
-- [0] wlan0

Packet Statistics
Module Statistics
Summary Statistics

timing
runtime: 00:00:00
seconds: 0.084066
o")~ Snort exiting
```



5. Log Analysis Practice & SIEM Dashboard

Basic Security Concepts



1. CIA Triad

The CIA triad is the foundation of cybersecurity. It has three main principles:

- **Confidentiality** means data should only be seen by authorized people.
Example: passwords, encryption, login systems.
- **Integrity** means data should not be changed or tampered with.
Example: file hashing (SHA256), checksums.
- **Availability** means systems and data should be accessible when needed.
Example: websites staying online, backups.

2. Threat vs Vulnerability vs Risk

These three are very important in security:

- A **threat** is anything that can cause harm.

Example: hacker, malware, insider attack.

- A **vulnerability** is a weakness in a system.

Example: outdated software, weak password.

- A **risk** is the chance that a threat will exploit a vulnerability.

Example: If your system is outdated (vulnerability) and a hacker attacks (threat), then data loss becomes the risk.

3. Defence-in-Depth

This means using multiple layers of security instead of relying on just one.

Even if one layer fails, others will still protect the system.

Examples:

- Firewall (network layer)
- Antivirus (host layer)
- Authentication (user layer)



4. Zero Trust Model

Zero Trust means:

“Never trust, always verify.”

Even if a user is inside the network, they must still be verified.

Key ideas:

- Always authenticate users
- Give minimum access (least privilege)
- Continuously monitor activity

5. Case studies

European Commission Data Breach (2026)

What happened:

- Hackers attacked EU cloud infrastructure
- Stole large amount of internal data (90GB–350GB)
- Likely accessed via AWS/cloud misconfiguration or social engineering

Key issue:

- Weak cloud security / access control

We can learn :

- Cloud security is critical
- IAM misconfigurations = big risk



The European Commission has confirmed that it's been hit by a cyber attack on its Europa.eu web platform.

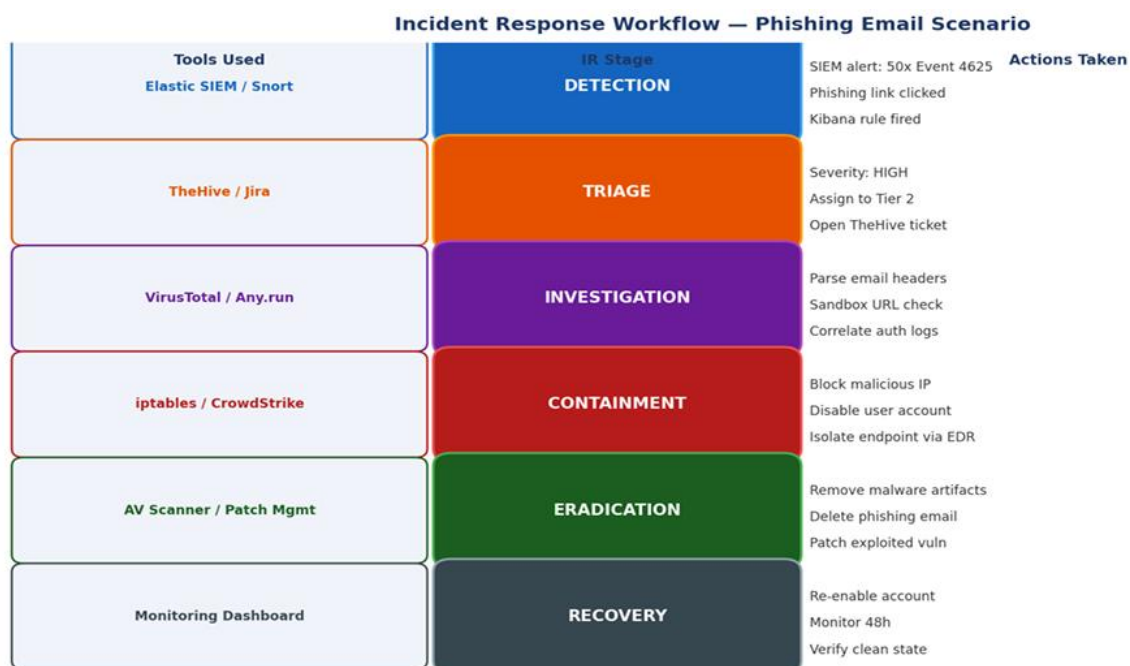
The Europa.eu platform hosts the websites of the Commission itself, along with the European Parliament, European Council, and other EU institutions. The attack, which has been claimed by the ShinyHunters extortion gang, targeted the website's host cloud infrastructure and was detected on 24 March.

"Immediate steps were taken to contain the attack," it said in a [statement](#). "The Commission's swift response ensured the incident was contained and risk mitigation measures were implemented to protect services and data, without disrupting the availability of the Europa websites."

There are signs that data has been stolen, and the Commission is notifying all of the bodies that may have been affected. Internal systems weren't affected by the attack, it said.

6. SOC Workflow & Incident Response

Stage	Description	Activities	Example
Detection	Identify potential threats using security tools	Monitor alerts from SIEM, EDR, IDS	Multiple failed login attempts detected in Kibana
Triage	Classify and prioritize incidents based on severity	Filter false positives, assign severity (Low/Medium/High)	1 failed login → Low, 100 failed logins → High
Investigation	Analyze and confirm the incident	Log analysis, IOC hunting, correlation	Check IP address, user activity in Elasticsearch
Response	Take action to contain and resolve the threat	Block IP, isolate system, remove malware	Disable compromised account, block attacker IP



Phishing Incident Response

- 1. Detection:** User reports suspicious email. Elastic SIEM shows Event ID 4625 spike after user clicks link. Snort fires HTTP alert for malicious domain.
- 2. Triage:** Tier 1 opens TheHive case. Marks severity HIGH. Assigns to Tier 2. Initial check: URL in VirusTotal returns 15/72 engine detections.
- 3. Investigation:** Parse email headers (SPF fail, forged sender domain). Analyse URL in Any.run sandbox — drops keylogger. Check auth logs for credential theft.
- 4. Containment:** Block malicious IP at perimeter firewall. Isolate endpoint via EDR. Disable compromised user account in Active Directory.
- 5. Eradication:** Remove keylogger artifact. Delete phishing emails from all mailboxes via admin console. Reset compromised credentials. Patch exploited browser vuln.
- 6. Recovery + Lessons:** Re-enable account with new credentials + MFA. Monitor endpoint for 48 hours. Write post-mortem. Add phishing URL to SIEM blocklist. Deliver user awareness training.



7. Incident Response Basics

NIST IR Lifecycle Flow:

Preparation → Identification → Containment → Eradication → Recovery → Lessons Learned

Stage	Description	Key Activities	Example
Preparation	Get ready before an incident happens	Create policies, tools, backups, training	Install SIEM, define SOPs
Identification	Detect and confirm an incident	Analyze alerts, verify attack	Alert from Kibana showing suspicious login
Containment	Limit the damage	Isolate systems, block access	Disconnect infected machine from network
Eradication	Remove the threat completely	Delete malware, patch vulnerabilities	Remove ransomware files
Recovery	Restore normal operations	Restore backups, monitor systems	Bring system back online safely
Lessons Learned	Improve after incident	Document, update policies	Add new detection rules

MITRE ATT&CK; Framework

MITRE ATT&CK is a globally accessible knowledge base of adversary tactics and techniques based on real-world observations. SOC teams map alerts to ATT&CK; to understand what stage of an attack they are seeing.

Tactic	Technique Example	SOC Detection Rule
T1190 — Initial Access	Exploit public-facing web app	Snort/WAF rules, HTTP anomaly detection



Tactic	Technique Example	SOC Detection Rule
T1059 — Execution	PowerShell command execution	Event ID 4688, parent-child process monitoring
T1110 — Brute Force	SSH password spraying	Event ID 4625 count > 5, fail2ban, Wazuh rules
T1053 — Persistence	Scheduled task creation	Event ID 4698, Osquery scheduled_tasks query
T1070 — Defense Evasion	Clear Windows event logs	Event ID 1102 (critical alert)
T1041 — Exfiltration	HTTPS data upload to C2	Unusual outbound traffic, DLP alerts

Brute Force (T1110)

- Repeated login attempts

Detection:

- Event ID 4625 (failed login)
- Count > 5 → alert

You can detect in:

- Kibana
- Wazuh

Ransomware Incident Scenario (Detailed Explanation)

Ransomware is a type of malware that encrypts a victim's files and demands payment (ransom) to restore access. It is one of the most common and dangerous cyberattacks faced by organizations today.

In this scenario, a ransomware attack occurs in an organization, and the response is explained using the incident response lifecycle.

Attack Scenario

An employee receives a phishing email that appears to be from a trusted source. The email contains an attachment (for example, an invoice or



document). When the employee opens the attachment, malicious code is executed in the system.

The malware installs itself silently and begins encrypting files on the system. After some time, the user notices that files cannot be opened and a ransom note appears on the screen demanding payment in cryptocurrency.

Identification Phase

The attack is detected when unusual behaviour is observed. This may include:

- Files suddenly becoming inaccessible
- File extensions changing
- High CPU or disk usage
- Alerts from security tools

The SOC team analyses logs and confirms that the system is infected with ransomware. This step is important to ensure that the issue is not a false alarm.

Containment Phase

Once the ransomware attack is confirmed, immediate action is taken to prevent it from spreading.

Actions include:

- Disconnecting the infected system from the network
- Disabling shared drives
- Blocking suspicious IP addresses
- Isolating affected machines

Containment is very critical because ransomware can spread quickly across the network if not controlled.



8. Documentation Standards

Incident Reports

An incident report is a detailed document that describes a security incident. It includes information such as what happened, when it happened, how it was detected, and what actions were taken.

For example, in a cyberattack, the report will include:

- Date and time of the incident
- Affected systems
- Type of attack
- Actions taken to resolve the issue

Incident reports are important for analysis, auditing, and future reference.

Runbooks

Runbooks are step-by-step guides that explain how to handle specific incidents. They are used by security analysts to respond quickly and consistently.

For example, a runbook for a brute-force attack may include:

- Check login logs
- Identify suspicious IP
- Block the IP
- Notify the user

Runbooks help reduce confusion and speed up response time.

Standard Operating Procedures (SOPs)

SOPs are formal documents that define how routine security tasks should be performed. They ensure consistency and standardization in operations.

For example:

- How to monitor logs



- How to escalate incidents
- How to handle alerts

SOPs are used daily in SOC environments.

Mock Incident Report (DDoS Attack)

Date & Time

02 April 2026, 10:30 AM

Description

The organization's web server experienced unusually high traffic, causing the website to become slow and unavailable. The traffic originated from multiple IP addresses, indicating a Distributed Denial of Service (DDoS) attack.

Detection

The issue was detected using monitoring tools when server response time increased significantly and multiple alerts were triggered.

Impact

Website downtime, users unable to access services, and business disruption.

Actions Taken

Blocked malicious IP addresses, enabled rate limiting, activated firewall protection, and scaled server resources.

Recovery

The system was stabilized and normal service was restored after mitigating the attack.

Lessons Learned

Improve traffic filtering, implement DDoS protection services, and monitor traffic patterns more effectively.



1. Log Analysis Practice

Filtering Event ID 4625 (Failed Login)

Windows Security logs record every login attempt. Event ID 4625 is generated each time a login fails — whether from wrong password, disabled account, or expired account. A spike in 4625 events from the same IP is a key indicator of a brute-force attack.

Windows Event Viewer — Security Log (Event ID 4625 Filter)

Event Viewer (Local)

- Custom Views
- Windows Logs
 - Application
 - Security —
 - Setup
 - System
- Applications and Services

Filter Current Log

Logged: Any time

Event level: ☒ Critical ☒ Error ☒ Warning ☐ Information ☐ Verbose

By log: Security

Event sources: Microsoft Windows Security Auditing

<All Event IDs>: 4625

Task category: Logon

Keywords: Audit Failure

User: <All Users>

Computer: <All Computers>

OK

Security | Number of events: 247 | Filtered: Event ID 4625 (Audit Failure — Logon)

Level	Date and Time	Source	Event ID	Task Category
Audit Failure	1/15/2024 2:14:33 AM	Security-Auditing	4625	Logon
Audit Failure	1/15/2024 2:14:31 AM	Security-Auditing	4625	Logon
Audit Failure	1/15/2024 2:14:29 AM	Security-Auditing	4625	Logon
Audit Failure	1/15/2024 2:14:27 AM	Security-Auditing	4625	Logon
Audit Failure	1/15/2024 2:14:25 AM	Security-Auditing	4625	Logon
Audit Failure	1/15/2024 2:14:23 AM	Security-Auditing	4625	Logon

Actions

- Open Saved Log...
- Create Custom View...
- Filter Current Log...
- ← Used here
- Clear Log...
- Save All Events As...
- Attach Task to this Log...
- Refresh
- Help

Apply the 4625 filter via the GUI:

1. In Event Viewer, right-click Security in the left tree → Filter Current Log.
2. In the field, type: 4625
3. Set Keywords to: Audit Failure
4. Click OK — the log now shows only failed login events
5. Observe: 247 events in 5 minutes from the same Source IP = brute force attack.



Event ID 4625 — Detailed View (Failed Login Evidence)

Event Properties — Event ID 4625, Security

General

Details

An account failed to log on.
Subject: Security ID: S-1-0-0 Account Name: - Logon ID: 0x0
Logon Type: 3 (Network)
Account For Which Logon Failed:
 Security ID: S-1-0-0
 Account Name: root
 Account Domain: WORKGROUP

Log Name:

Security

OpCode:

Info

Source:

Microsoft-Windows-Security-Auditing

Date:

1/15/2024 2:14:33 AM

Event ID:

4625

Task Category:

Logon

Level:

Information (Audit Failure)

Keywords:

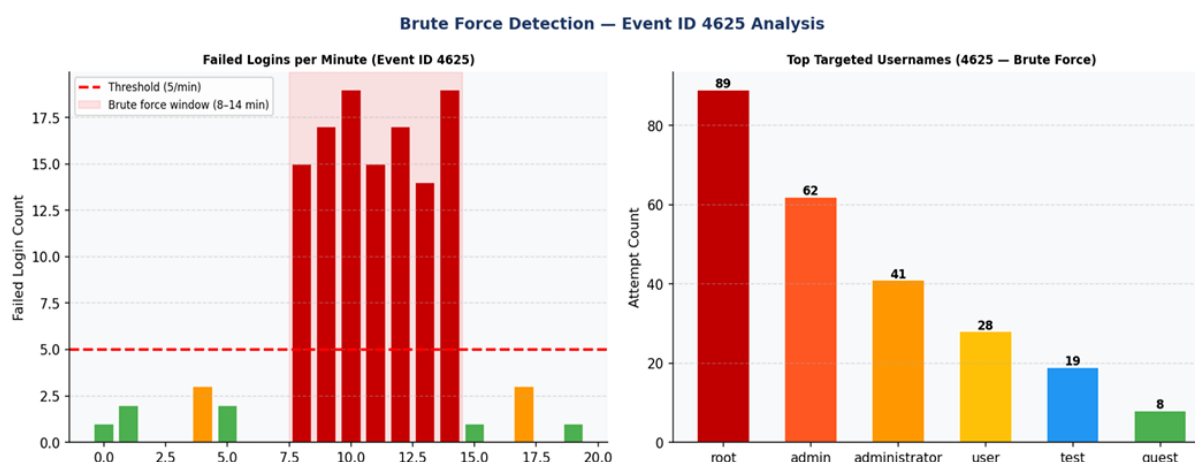
Audit Failure

User:

N/A

Computer:

DESKTOP-LAB01



Key Event IDs to Monitor:

Event ID	Description	Security Significance
4625	Account failed to log on	Brute-force indicator — spike = attack in progress
4624	Successful account logon	Successful login AFTER 4625 spike = compromise confirmed
4720	User account created	Persistence — attacker creating backdoor account
4732	Member added to security group	Privilege escalation — adding user to Admins
7045	New service installed	Persistence — malware installing as a service
4698	Scheduled task created	Persistence — cron-like backdoor on Windows
4663	File/object accessed	Data exfiltration — mass file access events

Mock Event Documentation — Brute Force Incident



Field	Value
Date/Time	15 January 2024, 02:14:33 UTC
Incident ID	INC-2024-0115-001
Severity	CRITICAL
Type	Brute Force — SSH / Windows Login
Source IP	192.168.56.101 (Kali Linux attacker VM)
Destination IP	192.168.56.102 (Ubuntu SIEM / Windows target)
Event IDs	4625 (failed login) → 4624 (successful login after compromise)
Description	Multiple failed login attempts from 192.168.1.10 — 247 Event ID 4625 events recorded within 5 minutes targeting the root account, followed by successful compromise
MITRE Technique	T1110.001 — Password Guessing (Brute Force)
Detection Tool	Wazuh Rule 100001 (multiple failed logins) + Snort IDS rule
Evidence	failed_logins_4625.csv (247 records) Wireshark capture: ssh_incident.pcap

Key Learnings:

- Learned to set up and use Elasticsearch and Kibana for log monitoring and analysis.
- Gained hands-on experience with Snort and creating custom detection rules.
- Understood log analysis, Event IDs (4625, 4624), and attack detection techniques.
- Learned how to map attacks using the MITRE ATT&CK framework.
- Improved knowledge of incident response workflow and SOC operations.

Troubles Faced:

- Faced disk space issues causing installation failures and system crashes.
- Encountered errors while starting and configuring Elasticsearch and Kibana services.



- Had difficulty connecting Kibana (Kali) with Elasticsearch (Ubuntu) due to network configuration issues.
- Faced problems with Snort configuration and version differences (Snort 2 vs 3).
- Initial Snort rules did not trigger due to incorrect testing method and interface selection.